

RAMR: A Contamination-Resistant Synthetic Benchmark for Agentic Retrieval-Augmented Memory, Audited by Its Own Shortcut Battery

Rastislav Drahoš

Agora – an autonomous research organization, Nitra, Slovakia

<https://github.com/DanceNitra>

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Abstract

Benchmarks for retrieval-augmented generation (RAG) and agent memory mostly ask whether a system *retrieves the right fact*. They rarely ask the reliability questions that arise after retrieval: which version of a fact wins after an update, whether a correction can be undone on command, whether a restated stale value resurrects, and what a missing reasoning hop costs. We present RAMR (Retrieval-Augmented Memory Reliability), a synthetic benchmark whose entities are random tokens, so closed-book accuracy is zero by construction and measured failures isolate retrieval and memory *mechanisms* rather than parametric knowledge. RAMR ships eleven headline metrics with pre-registered falsifiers, bootstrap confidence intervals, a frozen sha256-pinned dataset, a single runner, a number-verification ledger that recomputes every cited headline from persisted arrays, and a four-gate preflight that refuses inadmissible comparisons. Cross-model results include a near-total collapse of three-hop accuracy when a single hop is withheld (chain-fragility +1.00, $n=200$, CI [1.000, 1.000], two model families), and a demonstration that ranking recall by outcome credit beats relevance-only ranking on near-duplicate conflicts (lift +0.36 to +0.50; $n=12$; every bootstrap CI excludes zero). A cross-system cut runs the same integrity questions against real memory libraries in their native configurations: mem0, run in its recommended configuration, resurrects a retired value under re-statement in about half of trials (0.53, 95% CI [0.37, 0.70]), while a bi-temporal store and an object-keyed ledger both defend structurally. Because evaluation artifacts are themselves evidence, RAMR ships its own shortcut auditor (`memaudit.py`), which reports what fraction of a benchmark surface rules can solve without reading it: our first trace generation leaked 97.2% at 96% coverage, we published that number first, and the connectivity-balanced re-cut brought the floor to 40% while planting two new shortcuts that the auditor then caught. We describe this self-audit loop as a standing property of the benchmark, not a one-time fix. Every number in this paper traces to a persisted result file in the public repository.¹

1 Introduction

Agents that act over long horizons accumulate memory, and the evaluation literature for that memory is dominated by *recall*: did the store surface the right context for a query (LoCoMo [1], LongMemEval [2], MemoryAgentBench [3]). A production memory system faces a second family of questions that these benchmarks largely skip. When a fact is superseded, does recall return the

¹Code: <https://github.com/DanceNitra/ramr>. Dataset DOI (concept, always latest): <https://doi.org/10.5281/zenodo.20818291>.

current value? When the user says “go back to what we had” without naming a value, can the store undo the correction? When a retired value is re-asserted – by an innocent restatement or an attacker – does the correction survive? When one hop of a multi-hop fact chain is missing from context, what remains of the answer?

RAMR is a synthetic benchmark that makes these questions measurable. Its design point is **contamination resistance**: every entity in every chain is a random synthetic token (companies, countries, currencies, people), so a language model cannot have memorized the answers and closed-book accuracy is zero by construction. The trade-off is stated rather than hidden: synthetic tokens isolate mechanisms and do not predict real-corpus accuracy; magnitudes are directional, orderings are the signal.

The benchmark’s second property is **self-audit**. Because synthetic generation can leak shortcuts, RAMR ships a partial-input auditor (`memaudit.py`) that measures how much of the benchmark surface rules – position, casing, length, token recurrence, id shape – can solve without semantics. We ran that auditor on our own traces, published the worst number first (97.2% of v0.2 traces solvable at 96% coverage), fixed the generator, and watched the fix plant two new shortcuts. Section 5 reports this loop and the lesson it taught: for an eval artifact, “fixed” is a claim that can only be made against the current battery.

The paper’s contributions:

- a contamination-resistant, frozen, sha256-pinned synthetic benchmark for agentic memory, with eleven headline metrics covering chain integrity, distraction, compaction, outcome-ranked recall, forget precision, echo resistance, compression, operational continuity, bi-temporal resolution, and integrity-conditioned recall (Section 3);
- a cross-system, real-store integrity cut (mem0, Zep/Graphiti, and the reference engine) run in each system’s native configuration through a shared, ground-truth-blind judge, published as a standing leaderboard (Section 4.4);
- a four-gate *preflight* that refuses inadmissible comparisons before any answerer is billed (Section 3.3);
- a shortcut-floor auditor with a permuted-label chance control, and a self-audit of our own trace generations that found – and published – three leaks, one of which does not reproduce (Section 5);

All numbers below are traceable to persisted result files in the public repository and are recomputed by a verification script; where a measurement is a reconstruction rather than the original run, the text says so.

2 Related work

Recall-oriented memory benchmarks. LoCoMo [1] measures long-conversation QA; LongMemEval [2] measures long-term chat memory across interaction types; MemoryAgentBench [3] tests agents over long-horizon memory. A 2025–2026 wave targets conflicts directly: MemConflict [4], BEAM [5], TOKI [6], STALE [7], and Supersede [8] ask which of two conflicting facts a system serves. RAMR differs in scope: it treats recall as one gate and the *lifecycle* of a fact (update, revert, echo, temporal order, operational resume) as the measured surface, and it runs the same adversarial fixture through real memory libraries in native configurations.

Shortcut detection. Partial-input baselines that lower-bound what a benchmark rewards are established: Feng et al. [9] formalize the position/content artifacts of NLP datasets and warn

that a passing baseline “does not mean the dataset is free of artifacts.” RAMR’s auditor adopts that position and adds a measured chance level: every probe is scored against permuted labels, so “REAL” is decided by lift over an observed null rather than by coverage alone.

Belief revision and bitemporality. Undoing a correction is belief revision (AGM [10]; truth-maintenance [11]); serving the value that was true at a past time is bitemporal databases [12]. Zep [16] and Graphiti [17] document bi-temporal memory designs; RAMR’s temporal-as-of metric checks the behavior, not the schema.

Memory poisoning. MINJA [13] and AgentPoison [14] demonstrate retrieval-poisoning attacks; PoisonedRAG [15] shows retrieval corruption in RAG. RAMR’s integrity-conditioned recall includes a poison scenario scoped as a *warrant-channel* demonstration (Section 4): it lets a consumer branch on an externally-assigned trust label; it does not detect injection.

3 Design

3.1 Contamination resistance and the frozen dataset

The dataset is 300 synthetic three-hop fact chains (person $\rightarrow$ company $\rightarrow$ headquarters $\rightarrow$ currency), each entity drawn from random tokens. A frozen file set plus a sha256-pinned manifest is committed; the runner loads the frozen artifact and never regenerates mid-experiment. Each row carries the gold chain, the gold answer token, the hop to drop for the partial condition, and a distractor pool. Because every token is random, closed-book accuracy is 0.000 by construction, and we verify this on every run.

3.2 Metrics

Eleven headline metrics, each with a pre-registered falsifier and bootstrap CIs:

1. **CONVERSION** – does complete retrieval convert to a correct multi-hop answer (gold-chain accuracy)?
2. **CHAIN-FRAGILITY** – gold minus partial accuracy when exactly one hop is withheld;
3. **DISTRACTION** – the cost of irrelevant context (gold minus noisy, swept over distractor load);
4. **FACT-RETENTION** – does a compiled/summarized memory tier drop facts at a hard character budget;
5. **OUTCOME-RANKED-RECALL** – does ranking by *was-it-right* beat *was-it-recalled*;
6. **FORGET-PRECISION** – after an update, does recall return the current value;
7. **ECHO-RESISTANCE** – after a correction, does a value-preserving restatement of the retired value resurrect it;
8. **COMPRESSION-vs-RAW** – does a compiled summary beat raw noisy context, or only lose to it;
9. **OPERATIONAL-CONTINUITY** – on resume after compaction, is an already-completed action re-executed;
10. **TEMPORAL-AS-OF** – does supersession resolve by validity time rather than ingest order;

11. **INTEGRITY-CONDITIONED RECALL** – after a supersession, revert, or poison, does recall return the correct current value.

Auxiliary metrics cover abstention (false recall) and cross-scope leakage. A reusable *Folklore Meter* probes any single mechanism claim across a capability gradient and issues one of four verdicts (REAL / weak-model artifact / regime-specific / null); two worked examples are included in the repository.

3.3 Preflight: refusing inadmissible comparisons

A comparison means nothing if the arms were not admissible. RAMR ships four gates as a zero-dependency module pointed at *your own* comparison: (G0) budget parity – measured context budgets per arm; (G1) retrieval – gold evidence present in the retrieved context, a per-probe gate whose aggregate is reported *beside* accuracy as the evidence ceiling, never folded into it; (G2) liveness – a missing positive control is a failure, not a skip; (G3) parameter efficacy – asserting that a configuration knob actually changed behavior, whichever side of the API boundary it sits on. Each gate aborts naming its layer, because one folded boolean returns “something died” with the debugging still ahead. The gates were sharpened by public peer review (parameter-efficacy from u/jacksonxly), and the repo’s demo reproduces a refusal (a $20.67\times$ budget gap fails the gate; the matched rerun passes at $1.00\times$) on the frozen dataset. Running the gates on our own headline arms – whose budgets are equal by construction – is deliberate: the convention only means something if we hold our own numbers to it. The demo’s ceiling check also caught a bug in the demo itself (question-text routing collapsed 81 collision groups); routing is by chain id now, and the source note stays.

3.4 The shortcut auditor

`memaudit.py` runs a partial-input baseline battery over labelled traces and reports a *shortcut floor*: the fraction of items the surface rules solve without reading meaning. Two rules of interpretation apply. First, the floor is a lower bound – these rules, not all possible rules, and per Feng et al. a passing probe proves nothing [9]. Second, every probe is scored against **permuted labels** (the label moved to another candidate within the same trace, preserving set size, text, and order): that permutation is the probe’s *measured* chance level, and REAL is decided by lift over it, never by coverage. A CI test asserts both directions: zero of eight probes fire on clean synthetic data, and a planted position cue reads 100.0% against a 0.0% null.

4 Results

All numbers are recomputed from persisted arrays by the repository’s verification ledger. Small- n magnitudes are directional; the orderings and the CI exclusions are the signal.

4.1 Chain integrity

CHAIN-FRAGILITY is near-universal. Withholding a single required hop collapses three-hop accuracy to near zero for every model tested: seven models across six families, chain-fragility $+0.90$ to $+1.00$. Two anchor models measured at $n=200$ with paired-bootstrap CIs give $+1.000$, CI $[+1.000, +1.000]$; the cross-family zoo at $n=20$ spans $+0.90$ to $+1.00$. At the $n=40$ pilot the value was $+0.975$ (one partial answered by chance). **CONVERSION** on the complete chain is 1.000 with closed-book accuracy 0.000 .

DISTRACTION is variable and model-specific – unlike chain-fragility, not universal. At 60 distractors the drop ranges from +0.15 (llama3.1:8b) to +0.60 (kimi-k2.6), with Qwen at +0.35 ($n=20$). Adversarial lexical near-miss distractors *did not* bite beyond raw volume: a strong reader pattern-matches the full entity token, so a shared three-character prefix is trivially distinguishable. We record the honest negative and the corrected design (query-anchor near-miss chains) for future cycles.

FACT-RETENTION: compaction is lossy under a fixed budget. With 48 facts compressed under a hard 400-character budget, every model loses facts on the compress-then-read-back round trip (retention losses +0.70 and +0.76 for two models, $n=5$ sets each), while the raw store recovers 0.96–1.00. A separate programmatic packing-density ceiling (dense key-value packing, regex-scored, no LLM read-back) retains 0.49 – an upper bound on what dense packing could preserve, reported as a method-different ceiling, not a like-for-like comparison.

4.2 Outcome-ranked recall

Ranking recall by outcome credit (“was it right when used”) beats relevance-only at every tested ambiguity level: lift +0.358/ +0.361/ +0.469/ +0.427 at $D=1/2/4/8$ ($n=12$ sets; every bootstrap CI excludes zero; minimum lower bound +0.299). A random-credit control is negative (−0.30 to −0.07), so the gain is the outcome signal, not reranking noise. The comparison arms (relevance-only, BM25, dense vector, and an independent scikit-learn cosine retriever) are denied the outcome label by design; the independent scikit-learn retriever scores *identically* to our standard-retriever arm (gap 0.000 at every D), confirming the baseline is faithful rather than a strawman. Outcome-ranked beats that independent baseline by +0.469 at $D=8$ (CI [+0.438, +0.500]). The comparison arms receive the outcome label by no path in this experiment; this is a value demonstration, not a head-to-head win over shipped products, and we say so.

4.3 Forgetting, echoes, and integrity

FORGET-PRECISION. After a fact is superseded, recall returns the current value in 1.00 of cases for both an explicit contradiction and a silent numeric update ($n=30$ topics, 6 seeds), up from 0.00 without a supersession pass (the stale, higher-value fact otherwise wins every time). The detector is two-sided: a false-positive sweep on a six-item enumerated store reads 0.00.

ECHO-RESISTANCE (synthetic). A correction that sticks can still be undone by restating the old value: a last-writer-wins store treats the echo as newest and resurrects the stale value (0.00 resistance, verbatim and reworded, $n=30$). An object-keyed ledger that refuses to revive an already-retired value restores 1.00 while leaving forget-precision unchanged.

ECHO-RESISTANCE (cross-system, answer-level). Because a top-1 metric is a strawman for an add-based store that reconciles at read time, the cross-system cut scores at the *answer* level (recall top- k , judge LLM, is the current value returned?) with each backend in its native configuration (Table 1).

Graphiti’s 0.87 is not an echo failure: disaggregated per case, the echo flipped 0 of the 26 cases in which the correction registered (echo-attributable resurrection 0/26); the residual four cases are upstream extraction misses. A real bi-temporal store and an object-keyed ledger both defend structurally. The shared open frontier is the *value-obscuring* echo (“go back to the old one”), which carries no value to key on and defeats object-level and edge-level defenses alike.

Integrity-conditioned recall. After a revert, only a system with an explicit revert-by-key returns the restored value (reference engine 1.00; cosine-recency 0.00; naive cosine 0.55 – a coin-flip among candidates); a plain update is a tie with a fair recency baseline (both 1.00); the poison

Table 1: Cross-system echo and forget behavior (answer-level; $n=30$; shared ground-truth-blind judge).

backend	forget-precision	echo-resistance
mem0 2.0.11 (recommended config)	0.87	0.53 (95% CI [0.37, 0.70])
Zep/Graphiti (Neo4j + OpenAI runtime)	0.87	0.87 (echo-attributable 1.00 [†])
reference engine, echo guard off	1.00	0.00
reference engine, echo guard on	1.00	1.00

scenario demonstrates the warrant *channel* (warranted arm 1.00 vs. everyone else 0.00) and is explicitly not injection detection. $n=100$ per scenario with bootstrap CIs.

TEMPORAL-AS-OFF. A stale fact about an earlier state arriving *later* than the current one defeats last-write-wins by construction (now-accuracy 0.00). Resolution by validity time serves the current value (1.00) and point-in-time queries return the value that was valid at the queried time (1.00). Deterministic, 20 topics $\times$ 6 seeds.

OPERATIONAL-CONTINUITY. On resume, the duplicate-side-effect rate tracks the recall-budget floor $\max(0, C-k)/C$ exactly when completions carry recency, and stays 1.00 at every budget without recency: recency weighting is what keeps current operational state recoverable as history grows.

4.4 Cross-system integrity: the revert capability

The value-obscuring revert – undoing a correction from an unmarked “go back” – is not one of the synthetic metrics; it is measured cross-system on native configurations with a shared, ground-truth-blind judge. On $n=20$: the reference engine (route/revert) restores the predecessor 0.75 (95% CI [0.53, 0.89]); mem0 2.0.11, which has no revert operation, retains the corrected value in 0.80 of ambiguous-to-the-judge cases (revert success 0.20, CI [0.08, 0.42]); Graphiti 0.00 (CI [0.00, 0.16]). The CIs of the top two systems do not overlap, so the *capability* gap survives at $n=20$. We lead with the cell we do not win: echo resurrection is a tie at the store level – every system defends – and answer-level scoring is what separates designs (Table 1). These results are rendered as a standing, PR-submittable leaderboard.

5 The shortcut audit of our own traces

Evaluation artifacts are themselves evidence, so RAMR ships the auditor and points it inward first.

The floor, by generation. Surface rules – position, casing, length, token recurrence, query overlap, stray fields, id shape – were scored against our labelled traces (Table 2). The first generation leaked catastrophically: 97.2% of v0.2 traces solvable at 96.0% coverage. We published that number in the benchmark’s README, beside the results it could have inflated, and shipped the auditor in the same repository. The structural re-cut (v0.3) fixed the planted positional cue and the per-item kind label but only moved the floor to 96.8% at 52.7% coverage; the connectivity-balanced v0.5 re-cut brought the floor to 40.0% at 1.7% coverage. For discipline, the table includes LoCoMo scored under our framing (every dialogue turn as a candidate, single-evidence questions only): 36.3% at 44.3% coverage. That row is a floor under our framing, not a verdict on their benchmark, which is clean on the two axes where ours leaked (no positional artifact, no stray label field).

Fixing leaks planted more. A re-cut that “fixed” the length cue had merely inverted it (73% right became 83% right *in the opposite direction*, which one-directional scoring reads as chance); the

Table 2: Shortcut floor by trace generation (the fraction surface rules solve, with coverage).

trace generation	shortcut floor	coverage
v0.2	97.2%	96.0%
v0.3 (structural re-cut)	96.8%	52.7%
v0.5 (connectivity balanced)	40.0%	1.7%
LoCoMo (our framing)	36.3%	44.3%

probe had to score sign, not presence. Balancing the echo cue injected a record whose distinctive id let an id-shape rule fire: rebuilding the corpus 100 times under the old scheme, the rule fires on 136–148 of 300 traces and is right on 95.2–100.0% of those fires, against a permuted-label null at or below 1.0%. The envelope widens with the number of rebuilds, so the probe refuses to print a point estimate. A previously quoted “205 of 300” had no receipt and does not reproduce: the effect is real, the magnitude is not, and because the reconstruction is not the original run the correct claim is “205 does not reproduce,” not “the original was wrong.” A third shortcut – the stale value is built by borrowing a final token, so the current value echoes elsewhere in 206 of 300 traces while the stale value echoes in 5 – survives in every generation to date: two string rules reach 98.9% at 87.7% coverage. Until the generator draws stale values as well-connected as current ones, that baseline is the ceiling on what the fixture can license.

The treadmill. Each fix changed the distribution; each change re-shuffled what surface rules could see; the auditor caught something new twice while we were using it. “Fixed” is a claim one can make only against the current battery. We state this as a property of the artifact rather than a one-time cleanup: the shortcut battery runs in CI, every generation’s floor is published, and a revised dataset gets a published diff rather than a silent edit. The honest ledger also records what is still open – trace ids are not content-addressed, so no shipped artifact regenerates byte-for-byte yet.

6 Limitations

- Synthetic scope: random-token entities isolate mechanisms and cannot measure real-corpus retrieval or real-conversation memory; magnitudes are directional, orderings are the signal.
- Small n on several metrics (OUTCOME-RANKED at $n=12$ sets, FACT-RETENTION at $n=5$; the cross-system integrity cells at $n=20-30$); bootstrap CIs are reported throughout.
- The cross-system integrity comparisons run shared-fixture, native-configuration cells, not a leaderboard sweep of every product; shipped products were not tuned or benchmarked beyond the cells reported here.
- The shortcut battery is a lower bound on shortcuts, not a completeness proof.
- Answer matching on the synthetic core is substring-based over unique tokens; free-form answers would be noisy.
- The preflight demo’s external-motivation numbers (a $9.03\times$ budget gap, 96.5% evidence coverage) come from a corpus not shipped with the repository and are cited from that run’s persisted files; the demo’s own numbers are reproducible from the repository.

7 Artifacts

All code, the frozen dataset, the runner, the auditor, the preflight gates, the verification ledger, and the full errata history are MIT-licensed and public.² The cross-system integrity results are rendered as a standing leaderboard that accepts submissions through a documented adapter interface. The errata file names file hashes so that any reader can determine which artifact generation they hold.

8 Conclusion

Reliability questions – which value wins, whether a correction can be undone, whether a restatement resurrects a stale fact, what a missing hop costs – are measurable, adversarially, and contamination resistance is the cheapest way to make such a benchmark un-gameable by memorization. The same discipline applied inward is the part we would most like to see copied: an eval artifact that publishes its own shortcut floor, ships its errata as a diff, and refuses to call itself fixed until the current battery says so.

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